

Question 1 **Start a new page****Marks**

- (a) (i) The base of a certain solid S_1 is the region bounded by the parabola $y^2 = 4ax$ and the line $x = a$, where $a > 0$. 3

By taking slices parallel to the y -axis in this base where each cross-section is an equilateral triangle, find the volume of S_1 .

- (ii) The area bounded by $y^2 = 4ax$ and the line $x = a$ is rotated about the line $x = a$ to form a solid of revolution.

By considering slices parallel to the x -axis:

- (α) Show that the cross-sectional area A is given by: 2

$$A = \pi \left(a^2 - \frac{1}{2}y^2 + \frac{1}{16a^2}y^4 \right).$$

- (β) Hence, find the volume of the solid of revolution. 2

- (b) A particle P of mass m kg, is attached to the end of a light wire 5 cm long which rotates as a conical pendulum with uniform speed in a horizontal plane below a fixed point O to which the wire is attached. The particle rotates so that the angular velocity is ω rads/sec.

- (i) Show that the angular velocity is $\frac{26\pi}{5}$ rads/sec when the particle is rotating at 156 rpm. 1

- (ii) Find the semi-vertical angle θ of the conical pendulum (answer to the nearest degree and take $g = 9.8 \text{ m/s}^2$). 2

- (c) A particle moves in a straight line. It is placed at the origin O on the x -axis and is then released from rest.

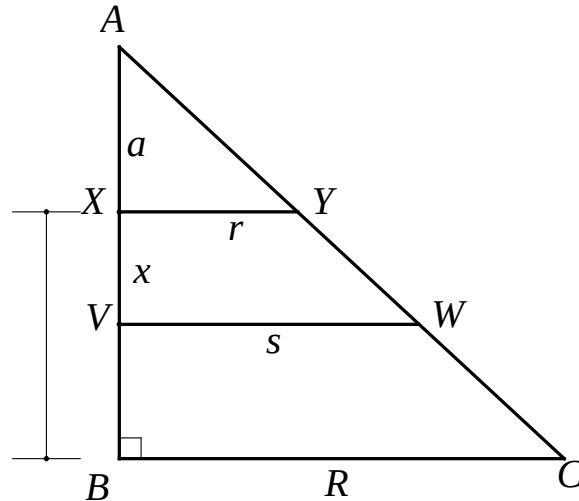
When it is at position x , the acceleration $\ddot{x}$, of the particle is given by:

$$\ddot{x} = -9x + \frac{5}{(2-x)^2}.$$

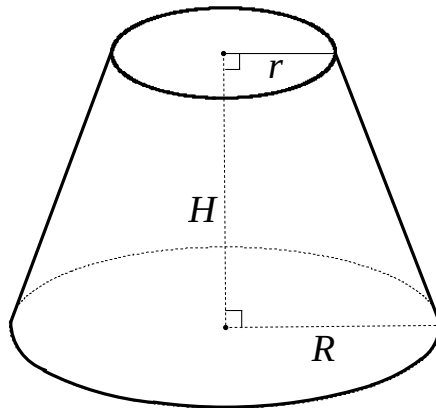
- (i) Show that: $v^2 = \frac{x(3x-5)(3x-1)}{2-x}$ for $x \neq 2$. 3

- (ii) Prove that the particle moves between two points on the x -axis, and find these points. 2

- (a) Consider the right-triangle ABC , where XY and BC are the lengths r and R respectively. Given VW is parallel to XY and BC . The distance between XY and BC is H , the length $VW = s$, $XV = x$ and length $AX = a$, as shown.



- (i) Show using similar triangles: $s = \frac{R-r}{H} \left(x + \frac{Hr}{R-r} \right)$. 3
- (ii) The frustum of a cone where the radius of the top and bottom faces are r and R respectively and the height is H , is shown below.



By considering a cross-sectional slice of the frustum parallel to the top face at a distance x units from the top, and using **integration**, show that the volume V of the frustum is given by :

3

$$V = \frac{\pi H}{3} (R^2 + Rr + r^2) .$$

Question 2 Continued

Marks

- (b) An object of unit mass falls under gravity through a resistive medium. The object falls from rest from a height of 50 metres above the ground. The resistive force, in Newtons, is of magnitude $\frac{1}{100}$ the square of the objects speed $v \text{ ms}^{-1}$ when it has travelled a distance x metres. Let g be the acceleration due to gravity in ms^{-2} .

- (i) Draw a diagram to show the forces acting on the body. **1**
Hence, show that the equation of motion of the body is:

$$\ddot{x} = g - \frac{v^2}{100}.$$

- (ii) Show that the terminal speed, $u \text{ ms}^{-1}$, of the body is given by: **1**

$$u = \sqrt{100g}.$$

- (iii) Prove that: **1**
 $\ddot{x} = v \frac{dv}{dx}.$

- (iv) Show that: **3**
 $\frac{v^2}{u^2} = 1 - e^{-\frac{x}{50}}.$

- (v) Find the distance fallen when the object has reached a speed equal to 50% of its terminal speed (correct to 1 decimal place). **2**

- (vi) Find the speed attained, as a percentage of the terminal speed, when the object hits the ground (correct to 1 decimal place). **1**

Question 3 **Start a new page**

Marks

- (a) The region bounded by the curves $y = \frac{1}{x+1}$ and $y = \frac{1}{x+2}$ and the lines $x = 0$ and $x = 2$, is rotated about the y -axis, forming a solid of revolution with a volume of V units³.

(i) Show that: $V = 2\pi \int_0^2 \frac{x}{(x+1)(x+2)} dx$. **2**

- (ii) Find V , correct to three significant figures. **3**

- (b) A vehicle is travelling along a horizontal straight road with a speed of 42 ms^{-1} . The engine is stopped as it passes a point marked O on the road and then the car is allowed to come to rest at a point B . The frictional resistance force is $\frac{1}{7}$ of the weight of the car and the air resistive force is $\frac{v}{14}$ per unit mass, where v is the speed of the car.

(i) If x is the distance travelled in metres, explain why $\ddot{x} = -\left(\frac{v+2g}{14}\right)$, **1**

where g is the acceleration due to gravity, in ms^{-2} .

- (ii) Find the distance travelled (to the nearest metre) **and** the exact time taken for the car to come to rest once the engine is stopped. **4**

Take $g = 10 \text{ ms}^{-2}$.

Question3 Continued

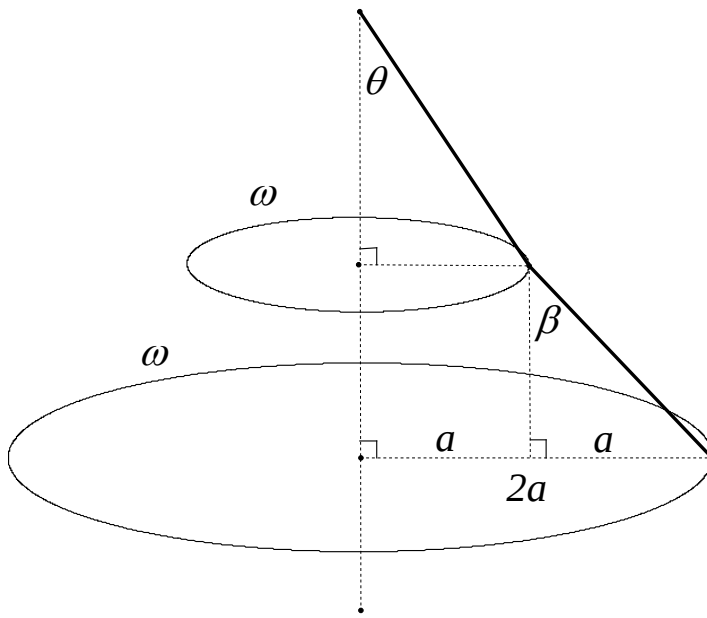
- (c) A light inextensible string ABC is such that $AB = \frac{5a}{3}$ and $BC = \frac{5a}{4}$.

A particle of mass m kg is attached to the string at C and another particle of mass $7m$ kg is fixed at B . The end A is tied to a fixed point and the whole system rotates steadily about the vertical AH (as shown), in such a way that B and C describe horizontal circles of radii a and $2a$ respectively and each has the same angular velocity ω .

- (i) By resolving the forces at C ,

show that the tension in the string BC is $\frac{5mg}{3}$ Newtons.

2



- (ii) Hence, find the tension in the part of string AB .

1

- (iii) Find the speed of the particle at B .

2

Question 4 **Start a new page**

Marks

- (a) A rectangular hyperbola has the equation $x^2 - y^2 = 8$.

Write down its eccentricity, the coordinates of the foci and the equation of each directrix.

Sketch the curve, indicating on your diagram each focus, directrix and asymptote.

5

- (b) This curve is rotated anti-clockwise through 45° , where the equation of the curve takes the form $xy = 4$.

- (i) Prove that the equation of the normal to the rectangular hyperbola $xy = 4$ at the point $P\left(2p, \frac{2}{p}\right)$ is $py - p^3x = 2(1 - p^4)$.

2

- (ii) If this normal meets the hyperbola again at $Q\left(2q, \frac{2}{q}\right)$ with parameter q , prove that $q = -\frac{1}{p^3}$.

2

- (iii) Hence, or otherwise, explain why there exists only one chord of the hyperbola where the gradients of the normal, at both ends, are equal.

Find the equation of this special chord PQ .

3

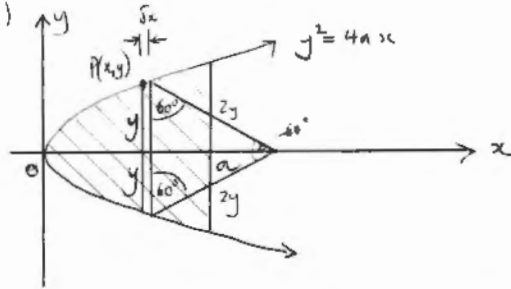
- (iv) Find the equation of the locus of the midpoint R of the chord PQ , as p and q vary.

3

End of Exam Paper

SOLUTIONS TO: YEAR 12 - Term 2 - ME 2 - 2008

Qn. ① (a) (i)



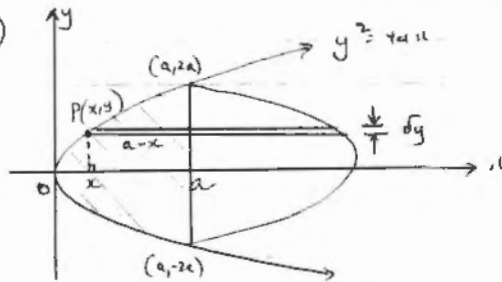
$$\begin{aligned} \text{Area of equilateral } A &= \frac{1}{2} \times 2y \times 2y \times \sin 60 \\ &= 2y^2 \times \frac{\sqrt{3}}{2} \\ &= \sqrt{3} y^2 \\ &= \sqrt{3} \times 4ax \quad (y^2 = 4ax) \\ \therefore A(x) &= 4\sqrt{3} ax \end{aligned}$$

Volume of cross-sectional slice = $\delta V = 4\sqrt{3} ax \delta x$

Volume of solid = $S_1 = \lim_{\delta x \rightarrow 0} \sum_{x=0}^a 4\sqrt{3} ax \delta x$

$$\begin{aligned} &= \int_0^a 4\sqrt{3} ax dx \\ &= 4\sqrt{3} a \left[\frac{x^2}{2} \right]_0^a \\ &= 4\sqrt{3} a \left(\frac{a^2}{2} - 0 \right) \\ &= 2\sqrt{3} a^3 \text{ units}^3 \end{aligned}$$

(ii)

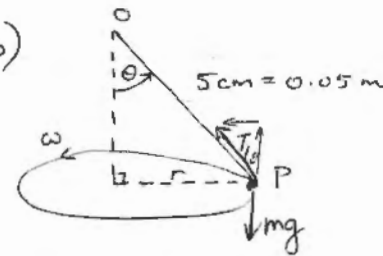


$$\begin{aligned} (a) \quad A &= \text{Area of cross-section} = \pi r^2 \\ &= \pi (a-x)^2 \\ &= \pi \left(a - \frac{y^2}{4a} \right)^2 \quad (\text{where } y^2 = 4ax) \\ &= \pi \left(a^2 - 2a \cdot \frac{y^2}{4a} + \left(\frac{y^2}{4a} \right)^2 \right) \\ \therefore A &= \pi \left(a^2 - \frac{1}{2} y^2 + \frac{1}{16a^2} y^4 \right) \end{aligned}$$

(b) Volume of cross-sectional disc = $\pi \left(a^2 - \frac{1}{2} y^2 + \frac{1}{16a^2} y^4 \right) \delta y$

$$\begin{aligned} V &= \pi \int_{-2a}^{2a} \left(a^2 - \frac{1}{2} y^2 + \frac{1}{16a^2} y^4 \right) dy \\ &= 2\pi \int_0^{2a} \left(a^2 - \frac{1}{2} y^2 + \frac{1}{16a^2} y^4 \right) dy \\ &= 2\pi \left[a^2 y - \frac{1}{6} y^3 + \frac{y^5}{80a^2} \right]_0^{2a} \\ &= 2\pi \left(2a^3 - \frac{1}{6} \times 8a^3 + \frac{32a^5}{80a^2} - 0 \right) \\ &= \frac{32\pi}{15} a^3 \end{aligned}$$

① (b)



$$\begin{aligned} (i) \quad \omega &= 156 \text{ revs/min} \\ &= 156 \times 2\pi \text{ radians/min} \\ &= \frac{156 \times 2\pi}{60} \text{ radians/sec} \\ \omega &= \frac{26\pi}{5} \text{ radians/sec} \end{aligned}$$

(ii) Resolving forces at P:

Vertically: $0 = mg - T \cos \theta$
 $mg = T \cos \theta \quad \text{--- (1)}$

Normally: $mr\omega^2 = T \sin \theta \quad \text{--- (2)}$

but $\sin \theta = \frac{r}{0.05} \Rightarrow r = 0.05 \sin \theta \quad \text{--- (3)}$

(2) ÷ (1): $\frac{T \sin \theta}{T \cos \theta} = \frac{mr\omega^2}{mg}$

$\tan \theta = \frac{r\omega^2}{g}$

$\therefore \tan \theta = \frac{0.05 \sin \theta \times \frac{26^2 \pi^2}{5^2}}{9.8} \quad \text{--- (4)}$ (From (3) and $\omega = \frac{26\pi}{5}$)

$\sec \theta = \frac{0.05 \times 26^2 \times \pi^2}{9.8 \times 5^2}$

OR $\cos \theta = \frac{9.8 \times 5^2}{0.05 \times 26^2 \times \pi^2}$

$\theta = 43^\circ \text{ (nearest degree)}$

$$1.(c) (1) \quad \ddot{x} = -9x + \frac{5}{(2-x)^2}$$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -9x + 5(2-x)^{-2}$$

$$\frac{1}{2}v^2 = -\frac{9x^2}{2} + 5(2-x)^{-1} + C \quad (1)$$

When $x=0, v=0$:

$$0 = 5(2)^{-1} + C$$

$$C = -\frac{5}{2}$$

$$\frac{1}{2}v^2 = \frac{5}{2-x} - \frac{9x^2}{2} - \frac{5}{2}$$

$$v^2 = \frac{10}{2-x} - 9x^2 - 5 \quad (1)$$

$$= \frac{10 - 9x^2(2-x) - 5(2-x)}{2-x}$$

$$= \frac{10 - 18x^2 + 9x^3 - 10 + 5x}{2-x} \quad (1)$$

$$= \frac{x(9x^2 - 18x + 5)}{2-x}$$

$$\therefore v^2 = \frac{x(3x-1)(3x-5)}{2-x} \quad [3]$$

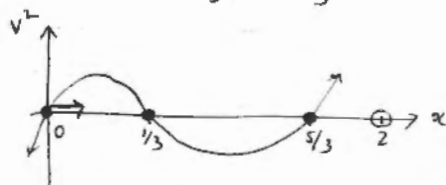
(ii) Particle is at rest ($v=0$):

$$0 = \frac{x(3x-1)(3x-5)}{2-x}$$

$$0 = x(3x-1)(3x-5), x \neq 2$$

$$x=0, x=\frac{1}{3}, x=\frac{5}{3}$$

at $x=0$,
 $\ddot{x} = \frac{5}{4}$
 $\therefore$ particle moving right.



$$v^2 > 0 \text{ when } 0 \leq x \leq \frac{1}{3} \text{ and } x \geq \frac{5}{3}, x \neq 2 \quad (1)$$

However, since particle was at $x=0$ initially, then the particle

3.

Qu 2

$$(a) (i) \quad \frac{a}{a+H} = \frac{r}{R} \quad (\text{corresponding sides of similar } \Delta\text{'s are in } \textcircled{1} \text{ proportion})$$

$$aR = ar + Hr$$

$$a = \frac{rH}{R-r} \quad \text{--- } \textcircled{1}$$

$$\text{Similarly: } \frac{a}{a+x} = \frac{r}{S}$$

$$aS = r(a+x)$$

$$S = \frac{r}{a}(a+x) \quad \text{--- } \textcircled{2} \quad (1)$$

$$\text{Sub. } \textcircled{1} \text{ into } \textcircled{2}: S = \frac{r}{\frac{rH}{R-r}} \left(\frac{rH}{R-r} + x \right)$$

$$\therefore S = \frac{R-r}{H} \left(\frac{rH}{R-r} + x \right) \text{ as required} \quad [3]$$

$$(ii) \text{ Volume of cross-sectional slice} = \pi s^2 \delta x$$

$$= \pi \left(\frac{R-r}{H} \right)^2 \left(\frac{rH}{R-r} + x \right)^2 \delta x$$

$$\text{Volume of solid} = V = \pi \left(\frac{R-r}{H} \right)^2 \int_0^H \left(\frac{rH}{R-r} + x \right)^2 dx \quad (1)$$

$$= \frac{\pi(R-r)^2}{H^2} \int_0^H \left(\frac{r^2 H^2}{(R-r)^2} + \frac{2xrH}{R-r} + x^2 \right) dx$$

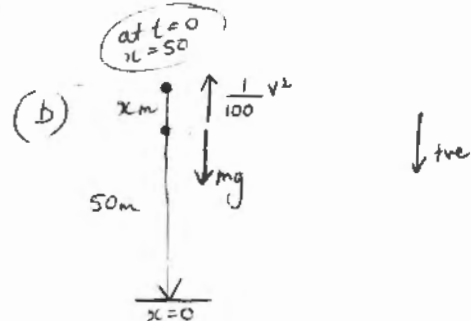
$$= \frac{\pi(R-r)^2}{H^2} \left[\frac{r^2 H^2 x}{(R-r)^2} + \frac{x^2 rH}{R-r} + \frac{x^3}{3} \right]_0^H$$

$$= \frac{\pi(R-r)^2}{H^2} \left(\frac{r^2 H^3}{(R-r)^2} + \frac{H^3 r}{R-r} + \frac{H^3}{3} \right) \quad (1)$$

$$= \frac{\pi(R-r)^2}{H^2} \cdot \frac{H^3}{3(R-r)^2} \left(3r^2 + 3r(R-r) + (R-r)^2 \right)$$

$$= \frac{\pi H}{3} \left(\cancel{3r^2} + 3rR - \cancel{3r^2} + R^2 - 2Rr + r^2 \right)$$

$$= \frac{\pi H}{3} (R^2 + Rr + r^2) \quad [3]$$



$$(i) \quad m\ddot{x} = mg - \frac{1}{100}v^2 \quad (1)$$

Since $m=1$

$$\ddot{x} = g - \frac{v^2}{100} \quad \text{as required} \quad [1]$$

$$(ii) \quad \text{Terminal velocity } (\ddot{x}=0)$$

$$0 = g - \frac{1}{100}u^2 \quad (1)$$

$$u^2 = 100g$$

$$u = \sqrt{100g} \quad [1]$$

$$(iii) \quad \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} \quad (1)$$

$$\ddot{x} = \frac{dv}{dx} \cdot v \quad \text{as required} \quad [1]$$

$$(iv) \quad v \frac{dv}{dx} = g - \frac{v^2}{100}$$

$$\frac{dv}{dx} = \frac{100g - v^2}{100v}$$

$$\frac{dx}{dv} = \frac{100v}{100g - v^2}$$

$$= 50 \left(\frac{2v}{100g - v^2} \right)$$

$$x = -50 \ln(100g - v^2) + C \quad (1)$$

$$\text{When } x=0, v=0, 100g = u^2 \Rightarrow$$

$$0 = -50 \ln u^2 + C$$

$$C = 50 \ln u^2$$

$$\therefore x = -50 \ln(u^2 - v^2) + 50 \ln u^2 \quad (1)$$

5.

$$-\frac{x}{50} = \ln \left(\frac{u^2 - v^2}{u^2} \right)$$

$$-\frac{x}{50} = \ln \left(1 - \frac{v^2}{u^2} \right)$$

$$1 - \frac{v^2}{u^2} = e^{-\frac{x}{50}}$$

$$\frac{v^2}{u^2} = 1 - e^{-\frac{x}{50}}$$

as required. (1)

[3]

$$(v) \quad v = \frac{1}{2}u$$

$$\frac{v^2}{u^2} = \frac{1}{4}$$

$$1 - e^{-\frac{x}{50}} = \frac{1}{4}$$

$$e^{-\frac{x}{50}} = \frac{3}{4}$$

$$-\frac{x}{50} = \ln \frac{3}{4}$$

$$x = -50 \ln \frac{3}{4}$$

$$\div 14.38410362$$

$$\therefore x = 14.4 \text{ (1 dp)} \quad (1)$$

[2]

$$(vi) \quad \text{Body hits ground, when } x=50:$$

$$\frac{v^2}{u^2} = 1 - e^{-1}$$

$$\frac{v}{u} = \sqrt{1 - e^{-1}}$$

$$= 0.795060097$$

$$\frac{v}{u} = 79.5\% \text{ (1 dp)} \quad (1)$$

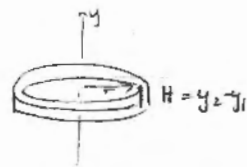
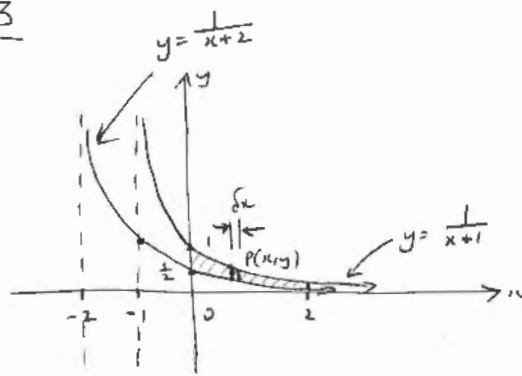
$$\frac{v}{u} = 79.5\% \times u$$

" Speed of particle when it hits the ground is 79.5% of terminal speed. [1]

6.

Qu 3

(a)



(i) By the method of cylindrical shells:

Volume of cylindrical shell = $\delta V = 2\pi x H$

$$= 2\pi x \left(\frac{1}{x+1} - \frac{1}{x+2} \right) \quad (1)$$

$$= 2\pi x \left(\frac{x+2 - (x+1)}{(x+1)(x+2)} \right)$$

$$= 2\pi x \times \frac{1}{(x+1)(x+2)} \quad (1)$$

$$\delta V = \frac{2\pi x}{(x+1)(x+2)}$$

$$\text{Volume of solid} = V = 2\pi \int_0^2 \frac{x}{(x+1)(x+2)} \quad (2)$$

(ii) Let $\frac{x}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$

$$x = A(x+2) + B(x+1)$$

Let $x = -1$: $-1 = A$

Let $x = -2$: $-2 = -B$

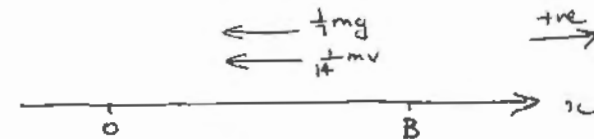
$$B = 2$$

} (1)

$$\begin{aligned} \therefore V &= 2\pi \int_0^2 \left(\frac{2}{x+2} - \frac{1}{x+1} \right) dx \\ &= 2\pi \left[2\ln(x+2) - \ln(x+1) \right]_0^2 \\ &= 2\pi (2\ln 4 - \ln 3 - 2\ln 2 + \ln 1) \\ &= 2\pi (2\ln 2 - \ln 3) = 1.81 (3 \text{ sf}) \text{ u}^3 \quad (1) \end{aligned} \quad (3)$$

7.

(b)



(i) $m\ddot{x} = -\frac{mg}{4} - \frac{mv}{14}$ (1)

$$\ddot{x} = -\left(\frac{v}{14} + \frac{g}{4} \right)$$

$$\dot{x} = -\left(\frac{v+2g}{14} \right) \quad (1)$$

(ii) Take $g=10$: $\dot{x} = -\left(\frac{v+20}{14} \right)$

$$\frac{dv}{dt} = -\left(\frac{v+20}{14} \right)$$

$$\frac{dt}{dv} = -\frac{14}{v+20}$$

$$t = -14 \int_{42}^v \frac{1}{v+20} dv$$

$$= -14 \left[\ln(v+20) \right]_{42}^v$$

$$= -14 \left[\ln(v+20) - \ln 62 \right]$$

$$t = 14 \ln \frac{62}{v+20} \quad \text{--- (A) (1)}$$

Time taken for particle to come to rest ($v=0$)

$$t = 14 \ln \frac{62}{20}$$

OR $t = 14 \ln 3.1 \quad (\div 15.84 \text{ sec}) \quad (1)$

From (A):

$$\frac{t}{14} = \ln \frac{62}{v+20}$$

$$\frac{62}{v+20} = e^{t/14}$$

$$62 e^{-t/14} = v+20$$

$$v = 62 e^{-t/14} - 20 \quad (1)$$

$$\frac{dx}{dt} = 62 e^{-t/14} - 20$$

$$x = \int_{14 \ln 3.1}^t (62 e^{-t/14} - 20) dt$$

$$= \left[-14 \times 62 e^{-t/14} - 20t \right]_{14 \ln 3.1}^t$$

9.

$$\therefore x = -868 e^{-\ln 3.1} - 20 \times 14 \ln 3.1 + 14 \times 62$$

$$= -868 \times \frac{10}{31} - 280 \ln 3.1 + 868$$

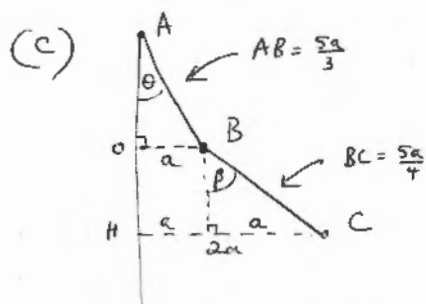
$$= 271.2074068$$

$$x = 271 \text{ m (nearest metre)}$$

①

$\therefore$ It takes $14 \ln 3.1$ sec and 271 m for the car to completely come to rest.

4



(i) At C, resolving forces:

Vertically:

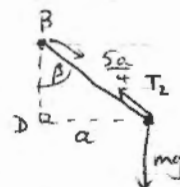
$$0 = mg - T_2 \cos \beta$$

$$T_2 = \frac{mg}{\cos \beta}$$

$$= \frac{mg}{\frac{3}{5}}$$

$$T_2 = \frac{5mg}{3}$$

①



$$BD = \sqrt{\left(\frac{5a}{4}\right)^2 - a^2}$$

$$= \sqrt{\frac{25a^2}{16} - a^2}$$

$$BD = \frac{3a}{4}$$

$$\therefore \cos \beta = \frac{3a/4}{5a/4} = \frac{3}{5}$$

$$\cos \beta = \frac{3}{5} \quad \text{①}$$

2

10.

(ii) Resolving forces at B:

Vertically:

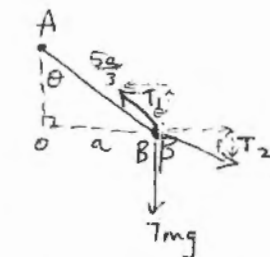
$$0 = 7mg - T_1 \cos \theta + T_2 \cos \beta$$

$$-7mg = T_2 \cos \beta - T_1 \cos \theta \quad \text{--- ②}$$

$$-7mg = \frac{5mg \times \frac{3}{5}}{3} - T_1 \times \frac{4}{5}$$

①

$$T_1 = 10mg$$



$$AO = \sqrt{\left(\frac{5a}{3}\right)^2 - a^2}$$

$$= \sqrt{\frac{25a^2}{9} - a^2}$$

$$= \sqrt{\frac{16a^2}{9}}$$

$$AO = \frac{4a}{3}$$

$$\therefore \cos \theta = \frac{4a/3}{5a/3} = \frac{4}{5}$$

$$\therefore \cos \theta = \frac{4}{5}$$

(iii) Need to find ω :

At C: $T_2 \cos \beta = mg$ (Vertically)
 $T_2 \sin \beta = 2am\omega^2$ (Horizontally)]

$$\frac{T_2 \sin \beta}{T_2 \cos \beta} = \frac{2am\omega^2}{mg}$$

$$\tan \beta = \frac{2a\omega^2}{g}$$

$$\omega^2 = \frac{\tan \beta \times g}{2a}$$

$$= \frac{4/3 g}{2a}$$

$$\omega = \sqrt{\frac{2g}{3a}} \quad \text{①}$$

At B, $v = r\omega = a \sqrt{\frac{2g}{3a}} = \sqrt{\frac{2ag}{3}}$

①

2

Qu 4

$$(a) x^2 - y^2 = 8 \Rightarrow \frac{x^2}{8} - \frac{y^2}{8} = 1 \quad \therefore a = \sqrt{8}, b = \sqrt{8}$$

$$\text{Using } b^2 = a^2(e^2 - 1) \text{ gives } 8 = 8(e^2 - 1)$$

$$1 = e^2 - 1$$

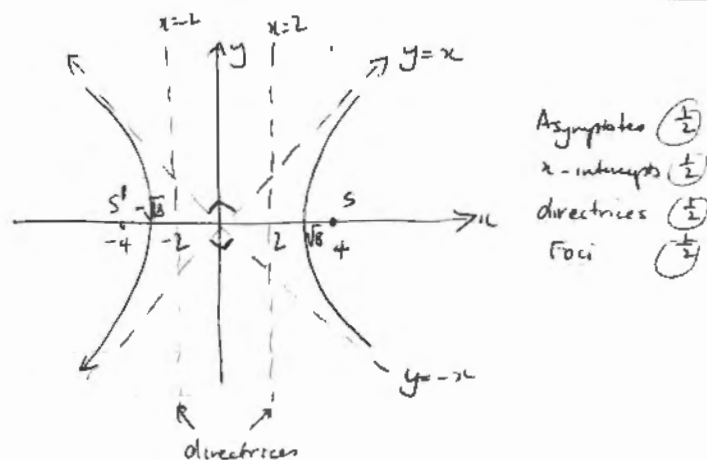
$$e = \sqrt{2}$$

①

$$\text{Foci are } S \text{ \& } S', \text{ using } (\pm ae, 0) = (\pm 4, 0). \quad \text{①}$$

$$\text{Eqns of directrices are } x = \pm \frac{a}{e} \Rightarrow x = \pm \frac{\sqrt{8}}{\sqrt{2}}$$

$$\text{ie } x = \pm 2 \quad \text{①}$$



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$$(b) (i) xy = 4 \Rightarrow y = 4x^{-1}$$

$$\frac{dy}{dx} = -4x^{-2}$$

$$\text{at } P(2p, \frac{2}{p}), \frac{dy}{dx} = \frac{-4}{(2p)^2} = \frac{-4}{4p^2} = -\frac{1}{p^2} \quad \text{①}$$

$$\left(\text{Alternative: } x \frac{dy}{dx} + y \cdot 1 = 0 \Rightarrow \frac{dy}{dx} = -\frac{y}{x} \right)$$

$$\text{at } P(2p, \frac{2}{p}), \frac{dy}{dx} = -\frac{\frac{2}{p}}{2p} = -\frac{1}{p^2}$$

$$\therefore \text{Gradient of normal} = p^2.$$

$$\text{Eqn of normal: } y - \frac{2}{p} = p^2(x - 2p)$$

$$py - 2 = p^3(x - 2p)$$

$$py - p^3x = 2(1 - p^4) \quad \text{①}$$

2

(ii) If normal passes through $Q(2q, \frac{2}{q})$, then it satisfies equation:

$$p \times \frac{2}{q} - p^3 \times 2q = 2(1 - p^4)$$

$$2p - 2p^3q^2 = 2q - 2p^4q \quad \text{①}$$

$$p - q = p^3q^2 - p^4q$$

$$p - q = p^3q(q - p) \quad \text{①}$$

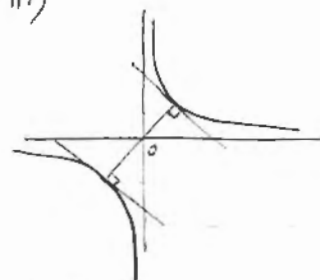
$$-1 = p^3q \quad (p \neq q)$$

$$\therefore q = -\frac{1}{p^3}$$

2

(iii)

From above, if PQ is a normal at P then $q = -\frac{1}{p^3}$ ie $p^3q = -1$.



Also, PQ is a normal at Q then $p = -\frac{1}{q^3}$ ie $pq^3 = -1$

$$\text{If PQ is a normal at both P and Q then } p^3q = pq^3$$

$$\text{ie } p^3q - pq^3 = 0 \text{ ie } pq(p^2 - q^2) = 0$$

$$\text{ie } pq(p - q)(p + q) = 0.$$

$$\text{Since } p \neq 0, q \neq 0 \text{ and } p \neq q \text{ then } \underline{p + q = 0} \text{ only.}$$

$$\text{ie } \underline{q = -p}$$

$$\text{Since } p^3q = -1, \text{ then } p^3(-p) = -1$$

$$p^4 = 1$$

$$p = \pm 1$$

$$\text{If } p = 1, \text{ Eqn of normal is } y - \frac{2}{1} = 1^2(x - 2 \cdot 1)$$

$$\text{ie } y = x$$

$$\text{If } p = -1, \text{ Eqn of normal is } -y + x = 2(1 - 1)$$

$$y = x$$

Thus, there is only one chord of the hyperbola where the gradients of the normal, at both ends, are equal. Its equation is $y = x$

(iv) Midpoint of PQ = R

$$\therefore R = \left(\frac{2p+2q}{2}, \frac{2p+2q}{2} \right)$$

$$R = \left(p+q, \frac{p+q}{pq} \right)$$

$$\begin{aligned} \therefore x &= p+q & \xrightarrow{\text{but}} & q = -\frac{1}{p^3} \\ \textcircled{1} \quad y &= \frac{p+q}{pq} & pq &= -\frac{1}{p^2} \quad \text{--- (A)} \end{aligned}$$

$$\frac{x}{y} = \frac{p+q}{\frac{p+q}{pq}}$$

$$= pq$$

$$\frac{x}{y} = -\frac{1}{p^2} \quad (\text{from (A)})$$

$$y = -\frac{1}{p^2 x} \quad \text{OR} \quad -\frac{y}{x} = p^2 \quad \textcircled{1}$$

Since R lies on the normal, it satisfies eqⁿ.

$$\text{i.e. } py - p^3x = 2(1-p^4)$$

$$y - p^2x = \frac{2}{p}(1-p^4)$$

$$-p^2x - p^2x = \frac{2}{p}(1-p^4)$$

$$-2p^2x = \frac{2}{p}(1-p^4)$$

$$-2x\left(-\frac{y}{x}\right) = \frac{2}{p}\left(1 - \frac{y^2}{x^2}\right)$$

$$2y = \frac{2}{p}\left(\frac{x^2-y^2}{x^2}\right)$$

$$4y^2 = \frac{4}{p^2}\left(\frac{x^2-y^2}{x^2}\right)^2 \quad (\text{squaring})$$

$$y^2 = \frac{1}{p^2}\left(\frac{x^2-y^2}{x^2}\right)^2 \quad \textcircled{1}$$

$$y^3 = -\frac{(x^2-y^2)^2}{x^3}$$

$$x^3y^3 + (x^2-y^2)^2 = 0$$

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